Organizational Conditions for AI-Driven Firm Performance

A Systematic Review of Empirical Evidence, 2015–2026

Bachelor's Thesis
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Abstract

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Chapter 1

Introduction

Chapter 2

Theoretical Background

2.1 Competitive advantage and firm performance

Competitive advantage is measured in 15 of the 67 studies reviewed here, and the remaining 52 report performance outcomes only. Barney separates the two constructs by what rivals are able to do: a firm holds a competitive advantage while it runs a value-creating strategy no rival, actual or potential, is running at the same time, and that advantage counts as sustained only once those rivals also cannot reproduce what the strategy yields (Barney, 1991, p. 102)[1]. Sustained in that sense says nothing about elapsed time. It asks whether the advantage is still there after rivals have given up trying to copy it (Barney, 1991, p. 102)[2]. This review therefore keeps the two apart throughout, because a firm can raise its margins without holding anything a rival is unable to copy.

Where both constructs appear, the separation is visible in the survey items. Chatterjee et al. ask about rivals directly: one of their competitive-advantage items states that implementing AI-based customer relationship management "will help an organization to win over its competitor which has not yet implemented AI-CRM for B2B relationship management" (Chatterjee et al., 2021, p. 215)[3]. Their performance items stay inside the firm and ask whether the same system improves "operational efficiency" and speeds up decisions (Chatterjee et al., 2021, p. 215)[4].

[1][Barney (1991), p. 102] "In this article, a firm is said to have a *competitive advantage* when it is implementing a value creating strategy not simultaneously being implemented by any current or potential competitors. A firm is said to have a *sustained competitive advantage* when it is implementing a value creating strategy not simultaneously being implemented by any current or potential competitors *and* when these other firms are unable to duplicate the benefits of this strategy."

[2][Barney (1991), p. 102] "Rather, whether or not a competitive advantage is sustained depends upon the possibility of competitive duplication. Following Lippman and Rumelt (1982) and Rumelt (1984), a competitive advantage is sustained only if it continues to exist after efforts to duplicate that advantage have ceased."

[3][Chatterjee et al. (2021), p. 215] "I think that successful implementation of AI-CRM will help an organization to win over its competitor which has not yet implemented AI-CRM for B2B relationship management."

[4][Chatterjee et al. (2021), p. 215] "I believe that successful implementation of AI-CRM will help the organization to improve its operational efficiency." / "I think AI-CRM helps in quick decision making

Mehta et al. give competitive advantage a position between adoption and results: they test it as the channel through which AI adoption reaches performance, and predict a positive effect on performance in its own right (Mehta et al., 2025, p. 550)[5].

Not every study that names competitive advantage builds its own instrument. Hossain et al. measure sustained competitive advantage with items carried over from earlier work (Hossain et al., 2022, p. 247)[6]. The construct label alone says little about what a study measured; the items decide.

2.2 Artificial intelligence as a general-purpose technology

Economists file AI under general-purpose technologies (GPTs), and that classification already predicts an uneven payoff. A GPT earns the label because it makes “new and complementary production methods” available, which raise output only over time (Czarnitzki et al., 2023, p. 188)[7]. Those complements are the expensive part. Before output moves, a firm has to redesign how work is organised, retrain staff, rework its software and accumulate managerial experience, and because none of that accumulation is booked as an asset, measured productivity dips before it climbs (Brynjolfsson et al., 2021, pp. 333–334)[8]. The lag is years, not quarters.

Two studies in this corpus show what that looks like at firm level. On German innovation-survey panel data, Czarnitzki et al. measure a clear productivity gain from AI use, for the yes/no measure and for the intensity measure alike (Czarnitzki et al., 2023, p. 201)[9]. Fontanelli et al. ask a different question of French firms and find that AI use raises the *volatility* of productivity growth; the effect sits with firms that buy AI from outside providers, and it weakens where more of the payroll is made up of ICT engineers and technicians (Fontanelli et al., 2025, pp. 1–2)[10]. Across the two results, the reading

which helps the organization to improve its operational performance.”

[5][Mehta et al. (2025), p. 550] “H8. Competitive advantage has a significant and positive impact upon firm performance.” / “H9. Competitive advantage has a mediating influence on adoption of AI technologies and firm performance.”

[6][Hossain et al. (2022), p. 247] “sustained competitive advantage items were taken from Cao et al. (2019).”

[7][Czarnitzki et al. (2023), p. 188] “The main feature of general-purpose technology is to enable new and complementary production methods that may increase productivity over time (Bresnahan and Trajtenberg 1995; Bresnahan et al., 2002; Brynjolfsson and Hitt 2003; Cardona et al., 2013).”

[8][Brynjolfsson et al. (2021), pp. 333–334] “realizing their potential also requires large intangible investments and a fundamental rethinking of the organization of production itself. Firms must create new business processes, develop managerial experience, train workers, patch software, and build other intangibles. This raises productivity measurement issues because intangible investments are not readily tallied on a balance sheet or in the national accounts.”

[9][Czarnitzki et al. (2023), p. 201] “We found that employing AI technologies has a positive and significant impact on firm productivity. In particular, we showed that both the use of AI and the intensity with which firms exploit the potential of AI significantly increase both sales and value-added.”

[10][Fontanelli et al. (2025), pp. 1–2] “Our results show that heightened volatility is concentrated among

this review takes forward is that the AI investment is the constant and the complements are the variable.

2.3 The resource-based view and the imitability problem

Across the studies reviewed here, the resource-based view (RBV) is the recurring reason given for expecting AI to pay off. Barney sets four conditions on a resource before it can hold an advantage in place: it has to be valuable, rare among competitors and imperfectly imitable, and no substitute may do the same strategic job (Barney, 1991, pp. 105–106)[11]. Alwakid and Dahri restate the four conditions as the VRIN acronym (Alwakid & Dahri, 2025, p. 14)[12]. That is a shorthand Barney himself comes close to when he lists "rareness, inimitability, non-substitutability" as the characteristics that turn a firm attribute into a resource (Barney, 1991, p. 106)[13]. Hossain et al. put the same argument in terms of what a firm does with what it owns, and tie the long-run case to holdings rivals cannot match (Hossain et al., 2022, pp. 241–242)[14].

Purchased AI fails that test, and the RBV literature says so plainly. Krakowski et al. note that AI as a technology resource costs almost nothing to reproduce and that little stands in the way of copying it, so the RBV expects whatever advantage it substitutes for to erode (Krakowski et al., 2023, p. 1426)[15]. Their own evidence relocates the advantage rather than denying it: what resists imitation is the combination a firm builds, not either side of it (Krakowski et al., 2023, p. 1446)[16].

AI buyers, whereas firms that develop AI internally experience no such association. Finally, we find that the AI-volatility link among 'AI buyers' is mitigated in firms with a higher share of ICT engineers and technicians, suggesting that AI's successful integration requires complementary human capital."

[11][Barney (1991), pp. 105–106] "To have this potential, a firm resource must have four attributes: (a) it must be valuable, in the sense that it exploit opportunities and/or neutralizes threats in a firm's environment, (b) it must be rare among a firm's current and potential competition, (c) it must be imperfectly imitable, and (d) there cannot be strategically equivalent substitutes for this resource that are valuable but neither rare or imperfectly imitable." Typographical errors as in the original.

[12][Alwakid & Dahri (2025), p. 14] "Traditionally, RBV posits that firms achieve sustained competitive advantage by strategically deploying valuable, rare, inimitable, and non-substitutable (VRIN) resources (Barney, 1991; Wang & Zhang, 2025)."

[13][Barney (1991), p. 106] "Firm attributes may have the other characteristics that could qualify them as sources of competitive advantage (e.g., rareness, inimitability, non-substitutability), but these attributes only become resources when they exploit opportunities or neutralize threats in a firm's environment."

[14][Hossain et al. (2022), pp. 241–242] "The resource-based view (RBV) portrays how a firm can optimally use its resources, which are non-substitutable, unique and valuable for achieving sustained competitive advantage (Wernerfelt, 1984)."

[15][Krakowski et al. (2023), p. 1426] "When AI substitutes humans' cognitive capabilities, the RBV expects the advantage that these capabilities provided to erode (Peteraf & Bergen, 2003). This is because, as a technology resource, AI has close to zero marginal reproduction costs and few imitation barriers (Brynjolfsson & McAfee, 2014, p. 31)."

[16][Krakowski et al. (2023), p. 1446] "Our findings show that humans remain sources of sustainable advantage by combining and integrating their capabilities with machine capabilities into uniquely complementary bundles (Argyres & Zenger, 2012)." / "the locus of sustainable advantage is neither human

Two consequences follow for this review. The RBV speaks to competitive advantage and not to performance, so a study reporting higher margins has not thereby shown that anything is defensible against rivals; the separation of the two constructs in Section 2.1 is a theoretical requirement, not only a coding convention. And if the resource itself is imitable, the variation worth studying lies in what surrounds it, which is why the conditions under which AI pays off are treated here as the object of analysis rather than as controls.

2.4 Automation and augmentation

The same investment can be deployed in two ways, and the strategy literature keeps them apart. Raisch and Krakowski define automation as machines taking a task over from a human, and augmentation as humans and machines working on a task together (Raisch & Krakowski, 2021, p. 192)[17]. Their argument against the popular advice to prefer augmentation is that the two cannot be held apart in practice. For any single task at any single moment the firm picks one approach, and that choice makes the two contradictory; widen the frame to several tasks over several years and each turns out to produce the other, which makes them interdependent (Raisch & Krakowski, 2021, p. 195)[18].

Corpus studies pick the distinction up as a design choice with different returns. Zebec and Indihar Štemberger treat the two as the use cases through which AI adoption reaches performance, automation paying off in efficiency and augmentation in what people and machines manage jointly (Zebec & Indihar Štemberger, 2024, p. 7)[19]. Their model then returns no significant direct effect of process automation on process performance once tasks become complex, and the recommendation they draw from it is a boundary condition: automate what is routine, augment what is ambiguous (Zebec & Indihar Štemberger, 2024, pp. 16–17)[20]. Where the line falls is itself unstable. In the fragrance industry the machine took over part of the perfumers' problem solving and could not touch the part that depends on smelling a scent and judging how people will react to it (Krakowski et al.,

nor machine capabilities, but what humans do with these capabilities."

[17][Raisch & Krakowski (2021), p. 192] "Whereas automation implies that machines take over a human task, augmentation means that humans collaborate closely with machines to perform a task."

[18][Raisch & Krakowski (2021), p. 195] "Automation and augmentation are contradictory, because organizations choose either one or the other approach to address a given task at a specific point in time." / "If we increase our analysis's temporal scale (from one point in time to its evolution over time) and spatial scale (from one to multiple tasks), we comprehend that, in the management domain, the two AI applications are not only contradictory but also interdependent."

[19][Zebec & Indihar Štemberger (2024), p. 7] "Automation involves machines taking over tasks from humans, leading to performance gains via resource, cost and information-processing efficiencies. In contrast, augmentation involves humans collaborating with machines, enhancing both human and machine skills and productivity."

[20][Zebec & Indihar Štemberger (2024), pp. 16–17] "The results show that BPA has no significant direct effect on BPP for more complex tasks in terms of efficiency of execution or scalability." / "In summary, automate routine, well-structured tasks and augment complex, ambiguous ones."

2023, p. 1445)[21].

2.5 Situated AI

Kemp starts from the puzzle the two previous sections produce: AI is available to everyone, so buying it should not yield an advantage, yet firms plainly differ in what they get out of it. His answer is situated AI, which he defines as AI hemmed in by the firm's own experience, structure and relationships (Kemp, 2024, p. 618)[22]. Three properties of the technology stand in the way, and the first of them is the imitability problem from Section 2.3 under another name: AI is generic, explicit and myopic. Being generic puts AI on a level with general human capital, which competitors can acquire and put to work much as the adopter does (Kemp, 2024, p. 620)[23].

What a firm does control is the situating. Kemp names three activities: grounding, which picks the experiences a firm's own AI learns from; bounding, which works on the experiences feeding a competitor's AI; and recasting, which keeps algorithms and the routines around them adjusting to each other (Kemp, 2024, pp. 618–619)[24]. The vocabulary is conditional throughout. None of the three is a property of the technology, and each is something a firm may or may not do.

Empirical work in the corpus arrives at the same place from the data side. Shi et al., studying Chinese listed firms, attribute the spread in outcomes to intangible complements, to how AI is governed, and to managerial capability that fits one setting only (Shi et al., 2025, p. 1)[25]. Kemp's paper is conceptual, so what it offers are propositions rather than findings. They are the propositions this review is positioned to examine, because the conditions Kemp derives are the conditions the coded studies report.

[21] [Krakowski et al. (2023), p. 1445] "While machines substitute perfumers regarding certain problem solving activities, they are unable to match the perfumers' unique ability to smell fragrances and predict the human emotions they trigger. This example illustrates that AI is likely to substitute some, but not all of complex business tasks' activities."

[22] [Kemp (2024), p. 618] "The present paper begins to resolve this puzzle by developing a theory of situated AI—AI whose agency is circumscribed in a firm's experiential, structural, and relational systems."

[23] [Kemp (2024), p. 620] "The generic, explicit, and myopic nature of AI all endanger its firm-level benefits, but are uniquely difficult to surmount because AI is both a machine and agentic (Smith & Lewis, 2011). For example, AI's generic nature is akin to general human capital. General human capital cannot typically underly interfirm advantages because a firm's competitors can usually acquire and deploy that knowledge in ways that closely mimic corresponding actions of the focal firm."

[24] [Kemp (2024), pp. 618–619] "Grounding involves orchestrating which experiences one's AI will be allowed to learn from across the organization. Bounding involves efforts to shape the experiences anchoring a competitor's AI. Recasting involves orchestrating the continual adaptation of algorithms and their surrounding routines to enhance AI's alignment with interdependent activities in a firm."

[25] [Shi et al. (2025), p. 1] "Drawing on a comprehensive dataset of Chinese listed companies spanning 2010–2022, this study investigates how AI adoption influences enterprise value through the mediating role of data assets and the moderating effect of managerial capability." / "These mixed outcomes suggest that the value-creating capacity of AI depends on more than technological sophistication alone; it hinges on complementary intangible resources, effective governance structures, and context-specific managerial capabilities—factors that current empirical research has only partially explored."

Chapter 3

Method

Chapter 4

Results

This chapter reports the evidence in two steps. Section 4.1 maps the 67 included studies descriptively: where the evidence comes from and how it is produced. Section 4.2 then synthesizes the conditions under which firm-level AI investments translate into performance gains or competitive advantage. All counts, groupings, and figures in this chapter are the author's own compilation from the coding table; the full table is reported in the appendix.

4.1 Descriptive mapping of the evidence base

The search window ran from 2015 to 2026, yet the earliest included study dates from 2020 (Figure 4.1). Growth is concentrated at the end of that window: 44 of the 67 studies, two thirds of the corpus, appeared in 2025 and 2026 alone. The evidence base is young.

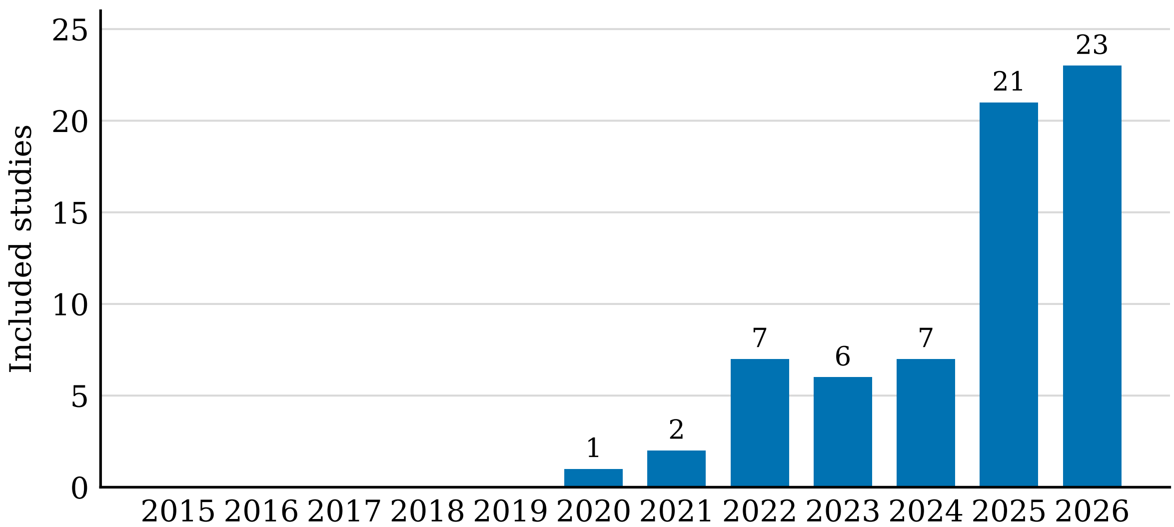

Figure 4.1: Included studies per publication year, search window 2015–2026 (n = 67). Source: own representation based on the coding table.

The 67 studies are spread across 45 journals, with no dominant outlet: Industrial Marketing Management and the Journal of Business Research lead with five studies each, followed by Technology in Society, the International Review of Financial Analysis, and the International Journal of Information Management with four each. Journal quality within the corpus is skewed toward the middle of the AJG scale (Figure 4.2): 27 studies come from journals rated 2, another 33 from journals rated 3, and seven from journals rated 4 or 4*.

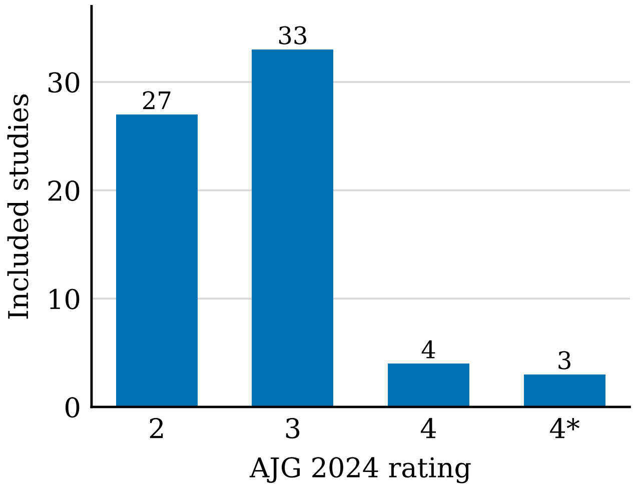

Figure 4.2: Included studies by AJG 2024 journal rating (n = 67). Source: own representation based on the coding table.

Two research designs dominate the corpus (Figure 4.3): panel econometrics on archival firm data (27 studies) and survey-based structural equation modeling (24 studies). The remaining sixteen studies combine several methods (six), examine stock market reactions in event studies (four), or build on case evidence (three); fsQCA, data envelopment analysis, and survey-based regression appear once each. The way studies operationalize AI investment splits the corpus in half: 33 studies rely on managers' self-reports through survey constructs, 31 on archival measures, and three on qualitative observation. Disclosure text mining (seven studies), patent-based proxies (six), workforce-based measures such as AI job postings (five), and investment announcements (four) account for most of the archival measures; the remaining nine rest on single-study instruments, for example procurement records or exposure to an AI policy shock.

Study samples cluster in two countries (Figure 4.4): the USA and China account for 16 studies each, together almost half of the corpus. India follows with seven studies, and another seven draw on multi-country samples. European firms appear in ten studies: four multi-country European samples plus six single-country ones. Most samples span several industries: 39 studies are cross-industry, ten focus on manufacturing, and the rest examine single settings such as healthcare or high-tech ventures.

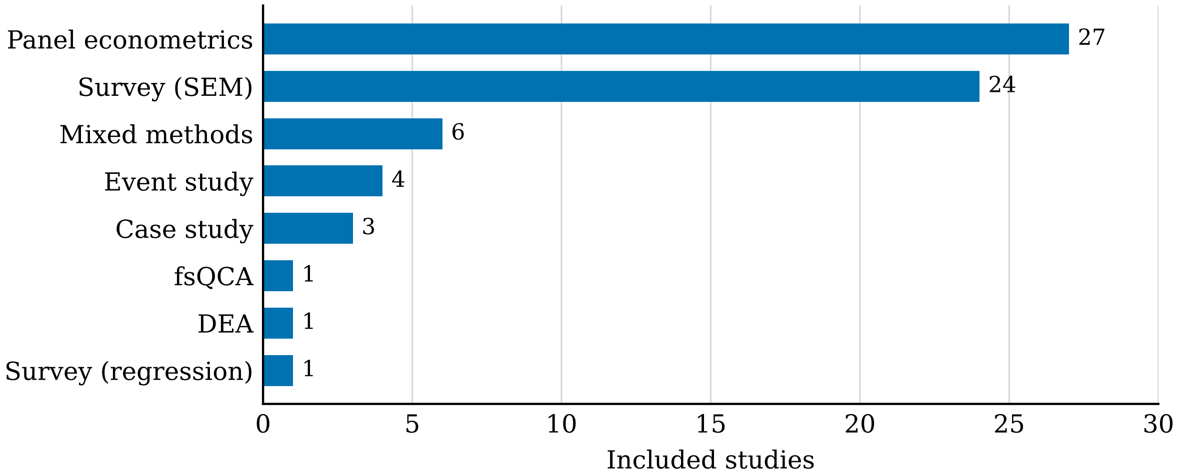

Figure 4.3: Included studies by research design (n = 67). Source: own representation based on the coding table.

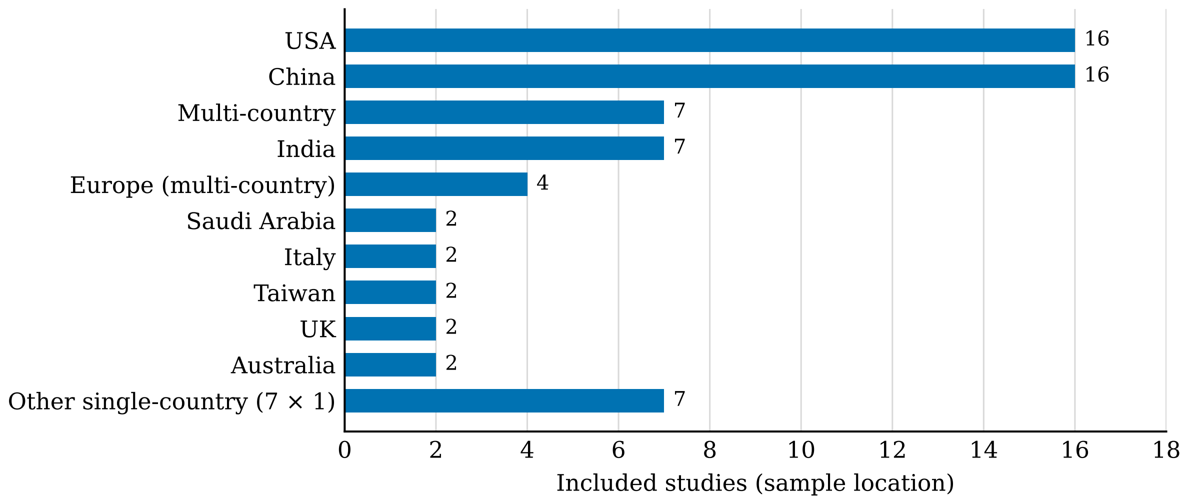

Figure 4.4: Included studies by sample location (n = 67). Source: own representation based on the coding table.

Following the distinction between superior firm performance and competitive advantage introduced in Chapter 2, the coding recorded which of the two constructs each study measures (Figure 4.5). The result is one-sided: 52 of the 67 studies measure firm performance only. Six measure competitive advantage as a distinct construct, and nine measure both. Measured competitive advantage is thus rare in this literature: 15 studies, under a quarter of the corpus. Effect directions spread across all four codes: 38 studies report a positive average effect of AI investment on their outcome, 20 an effect that exists only through a mechanism or under specific circumstances, five results that split across outcome indicators, and four negative effects. Among the 15 studies with a competitive-advantage construct, ten find positive and five conditional effects; none reports a negative one.

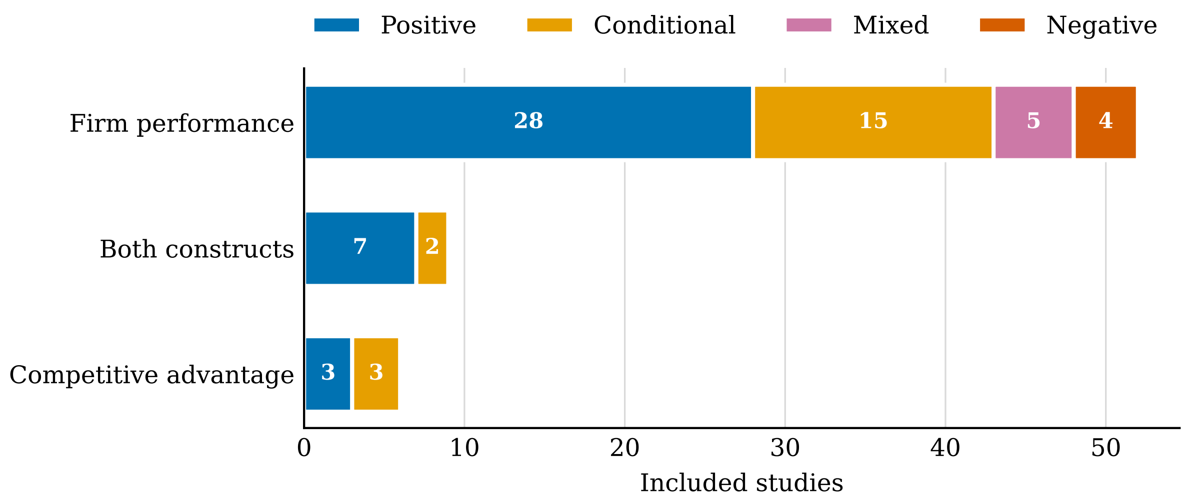

Figure 4.5: Outcome construct by reported effect direction (n = 67). Source: own representation based on the coding table.

Conditions are the rule, not the exception: 63 of the 67 studies identify at least one moderator, mediator, complement, or threshold that shapes the link between AI investment and its outcome. Only four studies report unconditional evidence. Section 4.2 orders these conditions.

4.2 Synthesis of conditions

Chapter 5

Discussion

Chapter 6

Conclusion and Future Outlook

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Appendix

Coding table

The full coding table for all 67 included studies, split into study characteristics (Table A.1) and findings (Table A.2).

Table A.1: Included studies: characteristics and design (n = 67). Source: author's compilation.

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S01	Wamba-Taguimdje et al. (2020)	Business Process Management Journal (2)	multi-country	~500 vendor-published AI project mini-cases (IBM, AWS, Cloudera, Nvidia etc.), archival qualitative analysis	case study	case observation (AI-based transformation projects from vendor websites)
S02	Baabdullah et al. (2021)	Industrial Marketing Management (3)	Saudi Arabia	392 B2B SMEs, cross-section, CB-SEM	survey-SEM	survey construct (acceptance of AI practices)
S03	Chatterjee et al. (2021)	Industrial Marketing Management (3)	India	349 managers from 16 BSE-listed firms (11 manufacturing, 5 service), response rate 51.6%, PLS-SEM	survey-SEM	survey construct (AI-CRM implementation)
S04	Chatterjee et al. (2022)	Journal of Business Research (3)	India	312 usable manager responses from 14 BSE-listed firms (10 large/4 SME; telecom, IT, automobile, retail, pharma), Jan-Feb 2020, RR 47.3%, PLS-SEM	survey-SEM	survey construct (adoption of AI-embedded CRM system)
S05	Hossain et al. (2022)	Industrial Marketing Management (3)	Bangladesh	257 usable manager responses (1,953 approached, 278 qualified), RMG export manufacturers, research-firm panel/Qualtrics, random sampling; time-lagged robustness subsample n=93	survey-SEM	survey construct (AI adoption on top of marketing analytics platform)
S06	Lee et al. (2022)	Technovation (3)	South Korea	160 high-tech ventures (from 1,248 ministry list, 300 responses), survey + audited financials	panel econometrics	survey (AI-adoption intensity) linked to financial records
S07	Leoni et al. (2022)	International Journal of Operations and Production Management (4)	Italy	120 senior executives of manufacturing firms, PLS-SEM	survey-SEM	survey construct (AI adoption/use)
S08	Lui et al. (2022)	Annals of Operations Research (3)	multi-market	119 AI investment announcements by 62 listed firms, Factiva search Jan 2015 - Feb 2019	event study	announcements (AI investment announcements)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S09	Mishra et al. (2022)	Journal of the Academy of Marketing Science (4*)	USA	19,000 firm-year observations, US-listed firms (COMPUSTAT + EDGAR 10-Ks), simultaneous equations; PSM robustness (1,801 treatment / 2,251 control firms)	panel econometrics	10-K text (share of AI keywords = AI focus)
S10	Sun et al. (2022)	Systems Research and Behavioral Science (2)	China	234 AI-related manufacturers (GM/CEO respondents), CB-SEM	survey-SEM	survey construct (value co-creation in AI innovation ecosystem)
S11	Ali et al. (2023)	Journal of Organizational Change Management (2)	UAE	single case (CMC Dubai): 9 semi-structured interviews with robotic-surgery team + archival data, triangulated	case study	case observation (adoption of AI/robotic surgery)
S12	Czarnitzki et al. (2023)	Journal of Economic Behavior and Organization (3)	Germany	5,851 firms (409 AI users, ~7%), German CIS 2018 (ZEW); cross-section OLS/2SLS-IV + panel subset with IV-FE, entropy balancing	panel econometrics	survey (CIS AI adoption dummy + intensity of AI use in business processes)
S13	Huang \{}& Lee (2023)	Data Base for Advances in Information Systems (2)	USA	109 AI implementation announcements 2014-2019, S&P 500 components (Compustat) + Google Search event collection	event study	announcements (AI implementation)
S14	Krakowski et al. (2023)	Strategic Management Journal (4*)	multi-country	same chess players across conventional, centaur and engine tournaments (controlled competitive setting)	panel econometrics	AI (engine) adoption in competitive interaction
S15	Samadhiya et al. (2023)	Industrial Marketing Management (3)	India	in-depth interviews with senior B2B managers + PLS-SEM on 142 responses	mixed	survey construct (AI-based partner relationship management implementation)
S16	Wu \{}& Monfort (2023)	Psychology and Marketing (3)	Taiwan	278 food firms (CEO/marketing managers), Taiwan, survey Mar-Aug 2020, RR 33%; SEM + supplementary fsQCA	survey-SEM	survey construct (implementation of AI marketing strategy)
S17	Babina et al. (2024)	Journal of Financial Economics (4*)	USA	1,993 US Compustat firms; resume-based AI workforce measure (Cognism), robust to job-postings measure; IV = university AI graduate supply	panel econometrics	resume-based (share of AI-skilled workers as firm AI investment)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S18	Cannas et al. (2024)	International Journal of Production Research (3)	Italy	multiple case study: 6 companies, 17 AI implementation cases, semi-structured interviews	case study	case observation (AI applications in SCOR processes)
S19	Pesqueira et al. (2024)	Computers and Industrial Engineering (2)	Europe	comparative 4 pharma companies (2 AI adopters vs 2 non-adopters) + 2-round Delphi with 53 participants	mixed	adoption status (AI in tender management)
S20	Pham et al. (2024)	Journal of Business Research (3)	USA	941 AI hospital-year obs + 941 matched non-AI hospital-year obs (246 matched hospitals), 40 US states, 2000-2020	panel econometrics	adoption dummy (hospital AI adoption)
S21	Sullivan \{}& Fosso (2024)	Journal of Business Research (3)	USA	two-wave survey, 107 US executives completing both waves (of 225 wave-1; 47.5% effective RR), panel-recruited IT+business executives	survey-SEM	survey constructs (AI-enabled automation, AI-enabled analytics, AI-enabled relational capabilities)
S22	Sun et al. (2024)	Technology in Society (2)	China	3,235 Chinese listed firms, 2007-2021; fixed effects + system GMM + mediation models	panel econometrics	annual-report text (frequency of AI-related terms); controls from CSMAR/RESSET
S23	Zebec \{}& Indihar (2024)	Business Process Management Journal (2)	EU	448 organisations, EU, serial multiple mediation SEM	survey-SEM	survey construct (AI adoption)
S24	Alam et al. (2025)	International Journal of Hospitality Management (3)	Malaysia	336 respondents, stratified purposive sampling, hospitality industry	survey-SEM	survey construct (AI adoption + technology readiness)
S25	Alwakid \{}& Dahri (2025)	Technology in Society (2)	Saudi Arabia	250 SMEs (managers/owners), manufacturing + services	survey-SEM	survey construct (AI capabilities: infrastructure, business integration, proactive stance)
S26	Banna \{}& Alam (2025)	International Journal of Entrepreneurial Behaviour and Research (3)	EU	unbalanced panel of 1,479 firms (˜12,900 obs), 26 EU countries, 2012-2023; CGM multi-way clustering + 2SLS-IV	panel econometrics	AI-related venture-capital funding received (log), OECD AI Policy Observatory data (turning point USD 11.3M); robustness: AI job intensity, AI patents, industry-level IV
S27	Basnet et al. (2025)	International Review of Financial Analysis (3)	USA	US firms, 10-K filings 2005-2018 (pre-hype window); AI mentions classified actionable/speculative/irrelevant; causal design	panel econometrics	10-K text (actionable vs speculative vs irrelevant AI disclosures)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S28	Bin-Nashwan \{}& Li (2025)	Technology in Society (2)	China	433 valid responses (461 screened, 5000+ invitations), Chinese accounting professionals, online survey Dec 2023-Jan 2024, PLS-SEM	survey-SEM	survey construct (AI-infused knowledge systems)
S29	Cao et al. (2025)	European Management Review (3)	USA	Study 1: instrument from 6,519 AI documents of 37 retailers; Study 2: fsQCA on survey of 140 U.S.-based retail managers	fsQCA	validated multi-level instrument (AI integration at task/function/firm level)
S30	Chakraborty (2025)	Current Issues in Tourism (2)	India	two-wave online survey, India: 1,487 invited, wave 1 n=437, wave 2 n=408 (hotel/resort managers, multi-channel recruitment)	survey-SEM	survey construct (GenAI adoption)
S31	Chiu et al. (2025)	Business Ethics the Environment and Responsibility (2)	Taiwan	33 publicly listed semiconductor firms; three-stage DEA (profit, sustainability, market) + Tobit regressions	DEA	patents (AI-driven innovation proxied by patent activity)
S32	D'Amico et al. (2025)	Journal of Technology Transfer (3)	UK	14,143 UK firms, 2004-2020	panel econometrics	archival: Beauhurst-identified AI technology/product use, matched to Business Structure Database + UK Innovation Survey (2004-2020, biennial)
S33	Feng et al. (2025)	Quarterly Review of Economics and Finance (2)	China	4,004 Chinese A-share firms, 2011-2023; procurement-based AI adoption index + patents + supply-chain data	panel econometrics	procurement index (AI adoption from procurement records)
S34	Fontanelli et al. (2025)	Journal of Economic Behavior and Organization (3)	France	French ICT survey 2019, firm-level; balancing/matching robustness	panel econometrics	survey (predictive AI use; buyers vs in-house developers)
S35	Guo et al. (2025)	International Marketing Review (3)	China	322 SMEs using cross-border platforms	survey-SEM	survey construct (AI adoption coupled with strategic agility)
S36	Huang \{}& Lin (2025)	Managerial and Decision Economics (2)	USA	S&P 500 firms: 50 AI adopters (23 first movers, 27 followers) identified via 97 manually screened AI-implementation news items + non-adopter controls; Compustat 2010-2017, baseline 2017	panel econometrics	news announcements (manually verified AI implementation)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S37	Kumar et al. (2025)	Journal of Business Research (3)	multi-country	Study 1: in-depth interviews; Study 2: SEM on 277 B2B managers across USA, UK, Canada, India, Australia, Malaysia, Japan	survey-SEM	survey construct (GenAI adoption)
S38	Mehta et al. (2025)	Journal of Business and Industrial Marketing (2)	India	Phase 1: 26 key-account-manager interviews (automobile); Phase 2: survey n=496, SEM	mixed	survey construct (adoption of AI technologies by key account managers)
S39	N. \{}& Roy (2025)	European Journal of Marketing (3)	Australia	335 companies across industries, PLS-SEM	survey-SEM	survey construct (AI-based stakeholder engagement)
S40	Sandeep et al. (2025)	Journal of Intellectual Capital (2)	India	290 HR professionals in talent acquisition, purposive sampling, South India, PLS-SEM	survey-SEM	survey construct (AI adoption in recruitment)
S41	Shi et al. (2025)	International Review of Financial Analysis (3)	China	Chinese listed companies 2010-2022 (~22,900 firm-year obs); Heckman correction, endogeneity tests	panel econometrics	annual-report text (Word2vec-based AI dictionary) + secondary measure
S42	Song et al. (2025)	Industrial Management and Data Systems (2)	USA	120 AI service failure events 2010-2024 at NYSE/NASDAQ-listed companies (from 451 initial events)	event study	announcements (AI service FAILURE events)
S43	Tao et al. (2025)	Journal of Business and Industrial Marketing (2)	China	291 B2B manufacturing firms, multi-wave multi-source, SEM	survey-SEM	survey construct (big data analytical intelligence assimilation + AI capabilities)
S44	Tingbani et al. (2025)	International Journal of Finance and Economics (3)	USA	1,950 unique US firms, 1996-2016, Compustat; system GMM	panel econometrics	resume/job-based AI workforce measure (Fedyk & Hodson 2023 approach)
S45	Adams et al. (2026)	Journal of Monetary Economics (4)	USA	US firms matched to Lightcast job postings 2010Q1-2024Q1 (40,000+ job boards); AI-pricing jobs identified	panel econometrics	job postings (AI-skill + pricing keyword = AI pricing adoption)
S46	Agag et al. (2026)	Tourism Management (4)	UK	1,206 UK listed tourism/travel/hospitality firms, 7,539 firm-years 2018-2024; two-step system GMM	panel econometrics	AI human capital (AI-skilled workforce measure)
S47	Alnofeli et al. (2026)	International Journal of Information Management (2)	Australia	3-stage: scoping review + expert interviews -> instrument; survey of 205 banking employees, SEM	survey-SEM	survey construct (AI-powered CRM capability, higher-order)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S48	Arshad et al. (2026)	International Journal of Information Management (2)	multi-country	two survey datasets of large firms worldwide (RPA + AI implementation)	survey-based regression (two cross-sectional surveys)	survey (joint implementation of RPA and AI)
S49	Bai et al. (2026)	Pacific Basin Finance Journal (2)	China	Chinese A-share listed firms; quasi-natural experiment (New-generation AI Pilot Zone Policy); pre-registered	panel econometrics	policy shock (AI pilot zone exposure) as exogenous AI investment driver
S50	Chen et al. (2026)	Journal of Empirical Finance (3)	USA	US firms; AI mentions in earnings calls, validated vs 10-K; IV = textual proximity to university AI research; 2SLS	panel econometrics	earnings-call text (share of AI terminology in management remarks)
S51	Dinh (2026)	Industrial Marketing Management (3)	Vietnam	sequential mixed-methods, 261 manufacturing SMEs	mixed	survey construct (AI-enabled knowledge capabilities)
S52	Filieri et al. (2026)	Journal of Product Innovation Management (4)	China	Chinese listed manufacturers + WTO tariff records + customs data 2015-2022; IV analysis	panel econometrics	archival AI integration measure (tariff-induced)
S53	Giordino et al. (2026)	Technological Forecasting and Social Change (3)	Europe	balanced panel, 432 European listed companies 2015-2023 (LSEG data), banks/insurers excluded	panel econometrics	organizational AI focus (archival, LSEG-based)
S54	Kazakis (2026)	Scottish Journal of Political Economy (2)	USA	US Compustat firms; efficiency via DEA (normalized ratio scores); labor-based AI investment measure	panel econometrics	labor-based AI investment measure
S55	Li et al. (2026)	International Journal of Production Economics (3)	China	218 firms in supply chains	survey-SEM	survey construct (responsible AI adoption)
S56	Li et al. (2026)	Journal of Business Research (3)	China	223 firms across industries, China, survey	survey-SEM	survey construct (GenAI affordances: organizational memory, collaborative, creational)
S57	Lin et al. (2026)	Emerging Markets Finance and Trade (2)	China	Chinese A-share listed companies, 2007-2022	panel econometrics	annual-report text-based AI adoption measure (authors explicitly contrast with patent/robot proxies)
S58	Lin et al. (2026)	International Review of Financial Analysis (3)	China	Chinese listed firms	panel econometrics	tailored AI-keyword dictionary, novel text analysis of company reports
S59	Liu et al. (2026)	International Review of Financial Analysis (3)	China	Chinese publicly listed firms, 2007-2024	panel econometrics	textual analysis of annual reports (CNINFO, jieba segmentation), ln(1+AI keyword frequency), following Mishra et al. 2022

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S60	Liu et al. (2026)	Technology in Society (2)	Asia-Pacific	Study 1: multilevel survey of 420 firms (HLM); Study 2: vignette experiment with 360 managers	mixed	survey construct (AI-enabled innovation); experimentally manipulated regulatory conditions
S61	Mukherjee et al. (2026)	International Journal of Quality and Reliability Management (2)	India	312 valid responses from luxury hotels (lodging + food services)	survey-SEM	survey construct (AI adoption)
S62	Patel (2026)	Technological Forecasting and Social Change (3)	USA	1,306 AI-scored service patents (of 758k patents) from 3,899 manufacturing firms, 1980-2020; matched-pair robustness	event study	patents (AI score of service technology patents)
S63	Renfei et al. (2026)	Technovation (3)	China	292 valid responses (295 distributed, two-stage collection; pretest 100/85), 120 Chinese manufacturing enterprises, PLS-SEM	survey-SEM	survey construct (AI capabilities: perceptive, predictive, prescriptive)
S64	Singh et al. (2026)	Business Strategy and the Environment (3)	USA	S&P 500 companies, 2000-2024; system-GMM + PSM robustness	panel econometrics	AI patents granted per year, identified via PATENTSCOPE AI Index methodology (1 SD = 115.66 patents -> Tobin Q +0.058, ~3% firm value)
S65	Song et al. (2026)	Finance Research Letters (2)	USA	US firms 2018-2023; BERT-classified AI disclosures vs AI patent portfolios	panel econometrics	text-vs-patents gap (unsubstantiated AI claims = AI washing)
S66	Ullah et al. (2026)	International Journal of Information Management (2)	China	419 manufacturing firms; SEM + LSTM validation	survey-SEM	survey construct (AI adoption)
S67	Wang \{}& Zhang (2026)	International Journal of Information Management (2)	China + Europe	two-wave survey of 357 early-stage ventures (China + Europe); PLS-SEM + IPMA + fsQCA + interviews	mixed	survey construct (AI innovation capability)

Table A.2: Included studies: outcomes, effect directions, and conditions (n = 67). Source: author's compilation.

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S01	Wamba-Taguimdje et al. (2020)	Perf.	Perf: organizational performance (financial, marketing, administrative) + process-level performance, qualitative	conditional	process reconfiguration required (mechanism: performance gains only when AI features are used to reconfigure processes); AI as configuration of complements (data, talent, domain knowledge, partnerships, infrastructure)	AI transformation projects improve organizational and process performance, but only when firms use AI to reconfigure their processes rather than overlay it on existing ones.
S02	Baabdullah et al. (2021)	Perf.	Perf: perceived SME performance (survey scale)	positive	technology roadmapping (enabler, sig.); attitude (sig.); infrastructure readiness (sig.); awareness (sig.); professional expertise and technicality NOT significant	Acceptance of AI practices improves perceived SME performance; adoption is driven by roadmapping, attitude, infrastructure and awareness - not by professional expertise.
S03	Chatterjee et al. (2021)	Both	Perf: perceived organizational performance (survey scale); CA: competitive advantage scale (COA1-3, Rogers-based; incl. market-share gain), discriminant-valid vs OP; predicted BY performance (H4: 0.59***, R2=.76)	positive	mediators: B2B engagement, employee experience, information processing capability (all sig.); moderator: leadership support (H5/H6 sig., MGA); firm size/age/industry only CONTROLS (not moderators)	AI-CRM implementation improves perceived organizational performance and competitive advantage in B2B relationship management.
S04	Chatterjee et al. (2022)	Perf.	Perf: firm performance scale (market+financial criteria, ROI/ROA, sales growth, market share, profitability) + B2B relationship satisfaction	positive	technology turbulence (negative moderator on ADM->BRS -0.23* and OPE->BRS -0.32***); leadership support (positive moderator on BRS->FP 0.29*); main effect ACRM->FP 0.53*** unconditional	AI-embedded CRM improves relationship satisfaction and firm performance; gains shrink under technology turbulence and grow with leadership support.
S05	Hossain et al. (2022)	CA	CA: sustained competitive advantage = RELATIVE performance vs competitors over last 3 years (market share growth, sales growth, profitability, ROI) - boundary case performance-as-CA, discuss in thesis	conditional	mediators: market sensing/seizing/reconfiguring (carry MAC->SCA fully, H3a-c); moderator: AI adoption strengthens MAC->sensing (0.104*), ->seizing (0.131*), ->reconfiguring (0.128*)	Marketing analytics capability drives sensing/seizing/reconfiguring toward sustained competitive advantage; AI adoption amplifies this only when built on an analytics foundation.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S06	Lee et al. (2022)	Perf.	Perf: revenue growth	conditional	AI-intensity threshold (low-level adoption: no effect); complementary technology investments (cloud, database) amplify; internal/exclusive R&D strategy amplifies	AI pays off only above an adoption-intensity threshold, and more so with complementary technology investments and venture-specific internal R&D.
S07	Leoni et al. (2022)	Perf.	Perf: perceived manufacturing firm performance (survey scale)	conditional	FULL mediation via knowledge management processes: direct AI->MFP 0.006 n.s. (H2 rejected), AI->KMPs 0.48***, KMPs->MFP 0.51***; also AI->SCR direct n.s.; firm size -> AI maturity (0.26**)	AI alone does NOT improve manufacturing firm performance - the effect exists exclusively through knowledge management processes (full mediation).
S08	Lui et al. (2022)	Perf.	Perf: firm market value (CAR; -1.77% on event day)	negative	nonmanufacturing firms (more negative); weak IT capability (more negative); low credit rating (more negative)	BENCHMARK: investors react negatively to AI investment announcements on average; the penalty concentrates in nonmanufacturing firms with weak IT capability and low credit ratings.
S09	Mishra et al. (2022)	Perf.	Perf: gross and net operating efficiency, net profitability, ad-spend, employment	mixed		AI focus reduces gross operating efficiency but increases net operating efficiency and profitability - automation raises costs more slowly than revenues.
S10	Sun et al. (2022)	CA	CA: perceived competitive advantage scale (own construct, Table 3) + innovation intelligibility; NO performance construct in model	conditional	FULL mediation: direct paths VC->CA and EV/RV->CA not significant; dynamic capabilities + innovation capabilities carry the effect (Leoni/Wang pattern: effect only via mechanism)	Value co-creation in AI ecosystems has NO direct effect on competitive advantage - effects run entirely through dynamic and innovation capabilities.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S11	Ali et al. (2023)	Both	Perf: four qualitative outcome categories: clinical (errors/infections down, patient experience), financial (quantified: ~500K/month at 30 robotic cases), organizational (resource allocation), technological; CA: competitive position with explicit competitor comparison: patient choice over competitors, bargaining power with third-party payers, differentiation, upper hand over competition (P1)	positive	qualified robotic team incl. revenue-cycle team (capability prerequisite); cultural shift in medical community; regulatory approval (DHA, case-by-case); high acquisition costs not covered by insurance (entry barrier)	AI-enabled robotic surgery improves clinical, financial and technological outcomes and strengthens the competitive position of the healthcare organization.
S12	Czarnitzki et al. (2023)	Perf.	Perf: firm productivity (labor- and value-added-based)	positive		AI adoption is robustly associated with higher firm productivity in representative German data; the effect grows with usage intensity.
S13	Huang \{}& Lee (2023)	Perf.	Perf: firm market value (positive abnormal returns on event day)	positive	announcement detail (detailed > vague); IT vs non-IT firms (sig. difference); frequency of negative words in announcement (negative correlation)	Markets reward AI implementation announcements on average - more so when announcements are detailed and come from IT firms.
S14	Krakowski et al. (2023)	Both	Perf: competitive performance (game outcomes/ratings); CA: sources of persistent performance heterogeneity (= sustained advantage) across settings	conditional	substitution: traditional human capabilities become obsolete; complementation: new human-machine capabilities emerge that are unrelated or negatively related to traditional ones	BENCHMARK: AI adoption shifts the sources of competitive advantage - a new decision-making resource emerges at the human-AI intersection, unrelated to traditional capability.
S15	Samadhiya et al. (2023)	Both	Perf: perceived firm performance (survey scale); CA: perceived sustainable competitiveness (survey scale)	positive	ICT capability (prerequisite); technological readiness (prerequisite); firm fit	AI-based partner relationship management improves firm performance and sustainable competitiveness, provided ICT capability and technological readiness are in place.
S16	Wu \{}& Monfort (2023)	Perf.	Perf: perceived firm performance (survey scale)	positive	fsQCA configurations: marketing capabilities + customer value co-creation + market orientation + AI strategy jointly form necessary and sufficient recipes for high performance	AI marketing strategy raises firm performance, but only as part of configurations with marketing capabilities, co-creation and market orientation - a configurational result.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S17	Babina et al. (2024)	Perf.	Perf: growth in sales, employment, and market valuation	positive	channel: product innovation (not process efficiency); gains concentrate among ex-ante LARGER firms -> rising industry concentration (superstar dynamics)	BENCHMARK: AI-investing firms grow in sales, employment and value via product innovation; benefits concentrate among large firms, increasing concentration.
S18	Cannas et al. (2024)	Perf.	Perf: qualitative competitiveness outcomes: costs, lead times, service level, quality, safety, sustainability	positive	barriers as boundary conditions: data quality, AI skills, high investment needs, unclear economic benefits, lack of AI cost-analysis experience	AI improves supply-chain competitiveness across SCOR processes, but realization depends on overcoming data-quality, skill and investment-evaluation barriers.
S19	Pesqueira et al. (2024)	Both	Perf: operational efficiency in tender management (processing times, decision quality, document accuracy) - expert-rated/qualitative; CA: expert-rated competitiveness dimensions per company (Delphi, Likert+IQR); bid competitiveness in tendering context	conditional	governance and ethical frameworks (contingency, per conclusions); regulatory compliance (EU AI Act, GDPR); organizational readiness; senior management involvement (identified critical factor); alignment with business goals	AI adopters manage pharmaceutical tendering faster and with better decisions than non-adopters, but only sustainably within robust governance and compliance structures.
S20	Pham et al. (2024)	Perf.	Perf: outpatient revenue, inpatient revenue, productivity, occupancy	positive	market share (larger-share hospitals adopt AI and benefit); endogeneity control essential - naive estimates misleading	Hospitals with larger market share adopt AI and gain in revenue, productivity and occupancy; without endogeneity controls the performance effects are misestimated.
S21	Sullivan \{}& Fosso (2024)	Perf.	Perf: firm performance + process innovation + product innovation (survey scales)	conditional	ARMC full mediator (no direct AI->performance paths modeled); environmental hostility (negative moderator on automation->ARMC -0.23*; relational path turns n.s.); environmental dynamism moderates; only 10/18 conditional indirect effects significant	AI capabilities pay off through the firms adaptive response to market changes; environment hostility/dynamism condition how much each AI capability contributes.
S22	Sun et al. (2024)	Perf.	Perf: firm total factor productivity	positive	stronger in state-controlled, internationally oriented, innovative firms; mechanism: reduced information asymmetry (key channel), specialized division of labor, green innovation; supply-chain digitalization policy amplifies; effect SLOWS over the long run	AI raises firm productivity in China - most for state-controlled, international and innovative firms - but the productivity boost decays over time.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S23	Zebec \{}& Indihar (2024)	Perf.	Perf: organizational performance (survey scale)	positive	serial mediation: decision-making quality and business-process performance carry the effect; process automation, organizational learning, process innovation as complementary partial mediators	AI creates business value indirectly: through better decisions and business-process performance, complemented by automation, learning and process innovation.
S24	Alam et al. (2025)	Both	Perf: value creation (survey scale); CA: sustainable competitive advantage (survey scale, modeled as mediator)	positive	SCA mediates AI adoption -> value creation; dynamic capabilities initially impede value creation (transition costs), later crucial under turbulence; technological turbulence moderates only the DC -> value creation path	AI adoption creates value in hospitality mainly through sustainable competitive advantage as transmission channel; dynamic capabilities help only once transition costs are absorbed.
S25	Alwakid \{}& Dahri (2025)	Perf.	Perf: sustainable SME performance (survey scale)	positive	SME creativity and green innovations (mediators); green entrepreneurial orientation as parallel driver (risk-taking, innovativeness, proactiveness)	AI capabilities raise sustainable SME performance through creativity and green innovation - AI infrastructure plus proactive strategy is the entry condition.
S26	Banna \{}& Alam (2025)	Perf.	Perf: firm revenue growth	conditional	U-SHAPE with QUANTIFIED turning point: ~USD 11.3M AI venture funding (Model 5: -0.679 linear, +0.140 quadratic) - below: revenue drag, above: gains; coupling with R&D innovation strategy amplifies (standalone AI insufficient)	AI investment follows a U-shaped payoff: short-term revenue drag, long-term gains - and only firms coupling AI with R&D strategy capture the upside.
S27	Basnet et al. (2025)	Perf.	Perf: firm value (market valuation); R&D spending, patents, lagged productivity as channels	conditional	disclosure SUBSTANCE: actionable disclosures -> valuation gains + innovation + lagged productivity; speculative/irrelevant -> nothing; silent or vague peers penalized	Markets reward only substantive (actionable) AI disclosures - early adopters with real implementation plans gain value via innovation; vague AI talk earns nothing.
S28	Bin-Nashwan \{}& Li (2025)	Perf.	Perf: accounting-firm performance: efficiency, accuracy, compliance, reporting innovation, timeliness (survey)	conditional	no direct AIK->performance path modeled; mediators: green human capital + green structural capital (carry effect), green RELATIONAL capital channel NULL; sustainability culture (moderator on GIC->performance)	AI-infused knowledge improves accounting-firm performance through green human and structural capital, amplified by sustainability culture; relational capital plays no role.
S29	Cao et al. (2025)	Perf.	Perf: efficiency and innovation outcomes (configurational)	conditional	CONFIGURATIONAL: no single AI function suffices - synergistic combinations (esp. customer service + cybersecurity with other functions); emergent capabilities: adaptive learning, predictive analytics, uncertainty mitigation	Retail performance comes from configurations of AI-enabled functions, not isolated AI uses - a directly configurational (fsQCA) result.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S30	Chakraborty (2025)	Both	Perf: firm/organisational performance (survey scales); CA: competitive advantage / competitive positioning (survey scale)	positive	technological readiness (TOE driver); leader narcissism moderates adoption-factor relationships and GAI effects on competitive positioning	GenAI adoption improves hotels competitive advantage and performance, with leadership personality (narcissism) shaping how adoption translates into outcomes.
S31	Chiu et al. (2025)	Perf.	Perf: stage efficiencies: profitability, ESG/sustainability, market valuation	mixed	stage-specific: AI innovation positive on SUSTAINABILITY + MARKET stage efficiency, insignificant on PROFIT stage (size/leverage drive profit); firm scale raises profit efficiency but LOWERS sustainability/market efficiency (asymmetry)	AI-driven innovation lifts sustainability and market-valuation efficiency in semiconductors but NOT short-term profit efficiency - the payoff is stage-specific.
S32	D'Amico et al. (2025)	Perf.	Perf: innovation performance (knowledge spillover of innovation)	conditional	complementarity with own R&D investment (AI pays only if it complements internal R&D); AI SUBSTITUTES knowledge collaboration with certain external partners	AI adoption boosts innovation only when paired with the firms own R&D investment - and partly replaces external knowledge collaboration.
S33	Feng et al. (2025)	Perf.	Perf: innovation output and innovation efficiency	positive	stronger for large incumbents and growth-stage firms; executives with IT backgrounds amplify; highly innovative firms diversify supply chains to reduce resource risk	AI adoption raises innovation output and efficiency, most where firm maturity, scale and IT-savvy leadership provide absorptive conditions.
S34	Fontanelli et al. (2025)	Perf.	Perf: VOLATILITY of productivity growth (not the level)	conditional	AI BUYERS: higher volatility; in-house DEVELOPERS: none; complementary human capital (share of ICT engineers/technicians) mitigates the buyer effect	Off-the-shelf AI raises the riskiness of productivity outcomes; developing in-house or holding complementary ICT human capital neutralizes it.
S35	Guo et al. (2025)	CA	CA: international advantage: product innovation + brand upgrading (GVC position, survey scales)	positive	international marketing capabilities (mediator); platform embeddedness DEPTH (inverted U on product innovation) and BREADTH (inverted U on brand upgrading) - medium embeddedness optimal	AI adoption coupled with strategic agility upgrades SMEs global value-chain position through marketing capabilities - but only at moderate platform embeddedness (inverted-U).
S36	Huang \{}& Lin (2025)	Perf.	Perf: financial ratios (Compustat), productivity, market value (Tobin q)	mixed	indicator-specific: financial performance + market value positive, productivity insignificant; first movers show partly negative/insignificant effects (no uniform first-mover advantage)	AI implementation lifts financial performance and market value, but first movers and top performers do not benefit uniformly across all indicators.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S37	Kumar et al. (2025)	Perf.	Perf: perceived firm performance (survey scale)	positive	ethical leadership (moderator of GenAI adoption -> performance); adoption reasons for/against (uniqueness, information completeness, convenience, deceptiveness)	GenAI adoption boosts perceived firm performance, more strongly under ethical leadership.
S38	Mehta et al. (2025)	Both	Perf: firm performance (survey scale); CA: competitive advantage (survey scale, modeled as mediator)	positive	manager personality traits (Big Five) drive adoption; competitive advantage mediates adoption -> performance; organizational culture moderates (agreeableness -> adoption)	Individual personality traits shape AI adoption in key account management, which converts into firm performance through competitive advantage - culture conditions the path.
S39	N. \{}& Roy (2025)	Perf.	Perf: firm performance (survey scale)	positive	marketing agility (mediator); technological turbulence AMPLIFIES the positive effects (high-turbulence environments gain more)	AI-based stakeholder engagement improves performance through marketing agility - and the payoff is largest in technologically turbulent environments.
S40	Sandeep et al. (2025)	CA	CA: perceived competitive advantage from AI-enabled recruitment (survey scale)	positive	HR competencies + open innovation -> dynamic capabilities (mediator); financial support + IT infrastructure as enabling resources	AI adoption in recruitment yields competitive advantage only where dynamic capabilities, built from HR competencies and open innovation with financial/IT support, are in place.
S41	Shi et al. (2025)	Perf.	Perf: enterprise value (Tobins Q)	positive	data assets mediate 45.6% (direct effect 0.0483*** remains); managerial capability moderates; boundary nulls: heavily polluting industries insignificant (unless credible green transition), asset-intensive firms null; SOE premium 0.035*** vs non-SOE 0.019**	AI adoption raises enterprise value through data assets, and only under capable management and agile (non-asset-heavy) structures - the study explicitly maps when AI becomes a source of advantage.
S42	Song et al. (2025)	Perf.	Perf: firm value (negative abnormal returns)	negative	small firms hit harder; recent years more negative; failure-type order effect: accuracy > safety > privacy > fairness	AI service failures destroy market value - most for small firms and accuracy failures; the downside risk of AI implementation is real and priced.
S43	Tao et al. (2025)	Perf.	Perf: new product performance (survey scale)	positive	AI capabilities (mediator: BDAI assimilation alone insufficient); electronic supply-chain collaboration (moderator, strengthens the indirect effect)	Big-data/AI assimilation lifts new product performance only when converted into AI capabilities, and more so with electronic supply-chain collaboration.
S44	Tingbani et al. (2025)	Perf.	Perf: firm growth (sales growth)	conditional	labour market conditions: labour productivity amplifies growth effect; labour cost and labour share weaken it (10% AI investment increase -> +0.04% growth on average)	AI investment raises firm growth modestly, and the payoff hinges on labour-market conditions - productivity-rich, labour-cost-light firms capture it.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S45	Adams et al. (2026)	Perf.	Perf: growth in sales, employment, assets, markups; stock-return response to monetary policy	positive	adoption concentrated in larger, more productive firms; adopters more responsive to monetary policy surprises	Firms adopting AI-powered pricing grow faster in sales, employment and markups - AI pricing is a capability large productive firms exploit first.
S46	Agag et al. (2026)	Perf.	Perf: firm value; wage inequality as mediating channel	positive	wage inequality partially mediates (AIHC compresses inequality -> value); stronger in dynamic/complex industries; sector split: value effect strongest in travel, inequality compression in hospitality	AI human capital raises firm value partly by compressing wage inequality - a workforce-centered pathway to AI-enabled advantage, strongest in dynamic sectors.
S47	Alnofeli et al. (2026)	Both	Perf: profitability (survey scale); CA: competitive advantage (survey scale)	positive	marketing ambidexterity (mediator: exploration/exploitation balance carries the effect)	AI-CRM capability lifts profitability and competitive advantage through marketing ambidexterity - capability without ambidexterity closes the value-realisation gap only partially.
S48	Arshad et al. (2026)	Perf.	Perf: revenue growth (via price increases); cost reduction tested	mixed	component split: joint RPA+AI -> additional REVENUE growth (via price increases), NO additional cost reduction (joint term n.s.; individual techs do reduce costs); strategic revenue-growth intent amplifies revenue outcomes	Combining RPA and AI pays through revenue enhancement, not cost efficiency - and only for firms whose strategic intent targets growth.
S49	Bai et al. (2026)	Perf.	Perf: stock price crash risk (up) + firm value (up)	mixed	crash risk rises via reduced information transparency + managerial optimism; worse for low-transparency and resource-constrained firms; firm value gains may be short-term market optimism	Policy-driven AI investment inflates firm value but simultaneously raises crash risk - short-term valuation gains trade off against long-term stability.
S50	Chen et al. (2026)	Perf.	Perf: corporate investment efficiency; Tobins q (long horizon)	positive	mechanisms: better sales forecasts, reporting quality, process/product innovation; data-privacy regulation exposure ATTENUATES the effect	AI adoption improves how firms allocate capital - unless regulatory friction (data-privacy exposure) blunts it; value shows up in Tobins q only over longer horizons.
S51	Dinh (2026)	Perf.	Perf: new product performance (survey scale)	positive	dual pathway: direct + via digital product innovation radicalness; digital sustainability leadership as BOTH mediator and moderator (reconfigures how AI creates value)	AI knowledge capabilities lift new product performance in emerging-market SMEs, with digital sustainability leadership as the critical boundary condition.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S52	Filieri et al. (2026)	Perf.	Perf: supply chain efficiency, sales performance, market value	conditional	INVERTED U: AI performance benefits diminish at extreme tariff exposure ("optimal investment threshold"); customer engagement needs strengthen the effect (boundary condition); tariff exposure induces AI integration	Trade tensions push firms into AI integration that improves efficiency, sales and value - up to a threshold where adversity overwhelms the compensation.
S53	Giordino et al. (2026)	Perf.	Perf: ROA + Tobins Q (+ ESG pillar scores)	positive		Organizational AI focus goes hand in hand with better financial performance AND better E/S sustainability scores in European listed firms.
S54	Kazakis (2026)	Perf.	Perf: firm efficiency	conditional	NO unconditional effect; gains only with: capable managers, stronger competitive pressure, stable institutional ownership, long-term debt access	AI investment alone does nothing for efficiency - gains materialize exclusively under managerial, competitive, ownership and financing conditions ("Conditional Gains").
S55	Li et al. (2026)	CA	CA: perceived competitive advantage (survey scale)	conditional	FULL mediation: direct responsible-AI -> CA path 0.091 n.s.; effect runs entirely via distributive + procedural justice; supply chain complexity weakens distributive path (-0.233***) but not procedural	Responsible AI builds competitive advantage through supply-chain justice - in complex networks, only procedural fairness remains a robust channel.
S56	Li et al. (2026)	CA	CA: perceived sustained competitive advantage (survey scale)	positive	three GenAI affordances mediate technological opportunism -> CA; competitive intensity amplifies ONLY the creational-affordance path	Technological opportunism converts into competitive advantage through GenAI affordances - under intense competition, only the creational affordance keeps paying.
S57	Lin et al. (2026)	Perf.	Perf: total factor productivity	positive	internal control quality (partial mediator: AI -> better governance/risk management -> TFP)	AI raises firm productivity partly by improving internal control quality - governance is a transmission channel, not just a moderator.
S58	Lin et al. (2026)	Perf.	Perf: investment efficiency + firm value	positive	stronger in high knowledge-intensity, high R&D-intensity firms and firms with sound internal controls; mechanisms: innovation capability + financial health	AI enhances investment efficiency and firm value most where knowledge intensity, R&D and internal controls provide absorptive capacity.
S59	Liu et al. (2026)	Perf.	Perf: M&A likelihood (strategic outcome) + long-term firm value	positive	stronger with abundant cash holdings, less-developed market environments, private ownership	AI adoption makes firms more acquisitive and raises long-term value - internal resources and institutional context set the boundary.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S60	Liu et al. (2026)	Perf.	Perf: productivity gains + firm growth	positive	regulatory climate moderates innovation->productivity link strength (clarity + supportive enforcement jointly maximize, moderated mediation); Study 2 experiment: causal evidence for regulatory conditions	AI-enabled innovation converts into productivity and growth only under clear, supportive regulation - institutional environment is the amplifier.
S61	Mukherjee et al. (2026)	Perf.	Perf: service performance (survey scale)	positive		AI adoption lifts hotel service performance where ease of use, competitive pressure, infrastructure and policy support jointly create momentum.
S62	Patel (2026)	Perf.	Perf: stock market reaction to patent grants (negative abnormal returns)	negative	idiosyncratic volatility mitigates the negative reaction; tangibility does not; effects only in 2010s/2020s (evolving market perception)	Markets react NEGATIVELY to AI-heavy service patents in manufacturing - a caution against assuming short-term market value from AI servitization.
S63	Renfei et al. (2026)	Perf.	Perf: corporate sustainability performance incl. explicit economic dimension (EP01-EP04: cost reduction, waste revenue, spin-offs)	positive	prescriptive capabilities FAIL (component of AI-capability bundle null); firm type boundary: Industry-4.0 manufacturers 0.58*** vs traditional significantly WEAKER (but not null)	AI capabilities raise sustainability (incl. economic) performance mainly in Industry-4.0-ready manufacturers - prescriptive AI fails, and traditional firms barely benefit.
S64	Singh et al. (2026)	Perf.	Perf: ROA + firm value	conditional	small positive ROA effect, stronger for R&D-intensive firms; firm-value gains ONLY for firms with good sustainability metrics; stronger under Democratic administrations (political environment as condition)	AI innovation pays modestly and unevenly: R&D intensity, sustainability credentials and even the political climate condition whether markets reward it.
S65	Song et al. (2026)	Perf.	Perf: market reactions + long-term operational performance	negative	time path: short-lived positive investor reaction, then negative BHAR + sustained underperformance; substantiation gap (claims vs patents) is the treatment	AI talk without AI substance backfires: markets initially reward the narrative, then punish rhetoric that outpaces implementation.
S66	Ullah et al. (2026)	Perf.	Perf: organizational performance (survey scale)	positive	team dynamics, corporate innovation capabilities, high-commitment workforce (mediators/amplifiers of the AI-performance link)	AI adoption converts into performance where cohesive teams, innovation capability and a committed workforce absorb it - human/organizational enablers unlock the value.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S67	Wang \{}& Zhang (2026)	Perf.	Perf: entrepreneurial success (venture performance, survey scale)	conditional	FULL mediation: strategic resilience fully accounts for AI capability -> success relationship (abstract: "fully accounts"); fsQCA equifinal configurations to high performance	AI capability makes ventures successful only via organizational resilience - the mechanism is organizational, not technological; multiple configurations lead there.

List of Aids

The following AI tools and aids were used during the preparation of this thesis.

Global (all sections)

Section	Tool	Description
ALL	Grammarly Pro	For mistake and style checking throughout the thesis. Mistakes were corrected immediately; style suggestions were reviewed manually and applied when suitable.
ALL	Claude Code (Anthropic)	Used throughout for LaTeX text correction and conversion, literature management (renaming PDFs, managing BibTeX), file structure, and drafting/editing section content. All AI-generated text was reviewed and approved before inclusion.

Section	Tool	Description
ALL (writing phase)	library-rag — local retrieval tool over the thesis corpus (built with Claude Code, 27 July 2026)	Local search system over the 67 full texts of the frozen corpus, used from the writing phase onward to locate citable passages. Runs entirely on the author's machine, no cloud services: Docling for PDF parsing, Qwen/Qwen3-Embedding-4B (fp8) for semantic search, BAAI/bge-reranker-v2-m3 for reranking, hybrid semantic and BM25 retrieval fused by reciprocal rank fusion. The tool retrieves and locates passages; it does not generate thesis text. Every verbatim quote it returns passes a two-stage programmatic check before it may enter a footnote: the quote must appear verbatim in the retrieved source passage, and its words must appear in the same order on the cited page of the PDF itself. The second stage also catches a quote attributed to the wrong passage or page. Its threshold was calibrated against 40 genuine and 60 fabricated quotes from this corpus (genuine: median 1.00; fabricated: maximum 0.18; the threshold of 0.60 admits none of the fabricated ones). Quotes the check cannot confirm are flagged for manual inspection of the page rather than passed through. Printed page numbers are calibrated per document from the printed page labels in the PDF and cross-checked against the Scopus page range; where no printed page number exists, the tool states this instead of supplying one. Corpus identification, screening, and coding were completed and frozen (17–18 July 2026) before this tool existed and are unaffected by it.
Proposal	Claude Code (Anthropic)	Readability rewrite pass over the proposal prose; minimal simplifications merged into proposal_final.tex after review; citations, quotes, and footnotes left unchanged.

Section	Tool	Description
Proposal	Quillbot Paraphraser	Comparison paraphrase of citation-free proposal sentences (proposal_quillbot.tex) to evaluate phrasing alternatives; citations, quotes, and footnotes left unchanged.
Proposal	Claude Code (Anthropic)	Source check of the proposal against the collected article PDFs: every cited claim compared with the source text, page numbers and verification footnotes added to all citations, and five wordings corrected where they had drifted from the source. All corrections reviewed before merging.
Proposal	Claude Code (Anthropic)	Shortening pass after I found the draft too dense: text condensed to 3 pages, one main statement per source, detail sentences removed. Citations and quotes kept verbatim; result reviewed by me.
Method	Claude Code (Anthropic)	Corpus retrieval (17 July 2026): wrote and ran the script corpus/pull_corpus.py, which executes the documented Scopus query via the Scopus API (WU license) and saves the frozen export (432 records, raw JSON + CSV). Verified that all three benchmark studies are retrieved and that record counts match Scopus.
Method	Claude Code (Anthropic)	Title/abstract pre-screening (17 July 2026): Claude read the full abstracts of all 174 studies passing the AJG ≥ 2 filter and proposed include/exclude decisions with reason codes against the criteria catalogue approved beforehand; borderline cases flagged for full-text checking. All proposals recorded in corpus/screening_2026-07-17.csv and reviewed by the author before becoming final.

Section	Tool	Description
Method	Claude Code (Anthropic)	Full-text screening assistance (17–18 July 2026): Claude extracted and summarized method/measurement sections from the PDFs of all flagged borderline studies and proposed verdicts against the criteria catalogue; 12 studies excluded content-based, 1 confirmed as include; two inaccessible reports excluded as not retrieved (author's decision). Claude also matched, renamed, and de-duplicated the downloaded PDFs. All verdicts recorded with reasons and reviewed by the author.
Results (coding)	Claude Code (Anthropic) + Gemini CLI (Google)	Dual independent AI coding of all 67 included studies against the frozen coding scheme (18 July 2026): Claude drafted all rows from the full texts (coder 1); Gemini 2.5 Pro coded the same studies blind—receiving only the coding scheme and PDF text, never Claude's drafts (coder 2). A documented self-consistency pass harmonized coder 1's effect-direction codes before comparison. Intercoder agreement: method 90%, outcome construct 88%, effect direction 73%.
Results (coding)	Claude Code (Anthropic)	Adjudication support (18 July 2026): for all 29 coder disagreements and 10 random-sample agreement rows, Claude performed full-text reads and prepared evidence briefs quoting both coders' positions with page references; 15 factual gaps resolved from the full texts. The author adjudicated all disagreements against these briefs (18 individually, the remainder in a final batch review); decisions recorded per case, decision rules documented as written case law. No joint coder errors found in the sample checks; all 67 rows finalized after the author's batch approval.

Section	Tool	Description
Theory (Section 2.1)	Claude Code (Anthropic) + library-rag	Drafting of Section 2.1 on the construct distinction (28 July 2026): citable passages located with the local retrieval tool; all four citations checked two-stage (verbatim in the source passage, word sequence on the cited PDF page) and their printed page numbers confirmed (pp. 215, 247, 550). Extended on 28 July 2026 by Barney's (1991, p. 102) definitions of competitive advantage and sustained competitive advantage, taken from the verified text sidecar and read off the page image; the corpus retrieval tool does not cover sources outside the corpus. The counts come from the fact sheet derived from the coding table, not from the tool. BibTeX entries generated from the frozen Scopus export with corpus/make_bib.py. Human-voice pass applied. Text reviewed and edited by the author before inclusion.
Results (descriptive mapping)	Claude Code (Anthropic)	Descriptive figures (18 July 2026): wrote and ran the script corpus/make_figures.py, which derives five corpus-mapping charts (publication year, method, sample geography, outcome construct $\times$ effect direction, AJG rating) directly from the finalized coding table and writes them to figures/ as PDF. Category groupings are documented in the script; all counts verified to sum to $n = 67$. Colors validated as colorblind-safe. Figures reviewed by the author before inclusion.

Section	Tool	Description
Theory (Sections 2.2–2.5)	Claude Code (Anthropic) + library-rag	Drafting of the four theory sections on general-purpose technology, the resource-based view, automation/augmentation and situated AI (28 July 2026). Citable passages from the corpus located with the local retrieval tool; all 11 corpus quotes checked two-stage (verbatim in the source passage, word sequence on the cited PDF page). Two quotes flagged by the check were inspected manually on the page; one revealed a page error (Krakowski et al., 1444 $\rightarrow$ 1445), which was corrected. The three theory anchors outside the corpus (Kemp 2024, Raisch & Krakowski 2021, Brynjolfsson et al. 2021) are not in the retrieval index; their passages were read directly from the PDFs with pdfplumber and the printed page numbers verified against the running heads. Barney (1991) is cited from the verified text sidecar to the image scan (literature/barney1991firm.txt, OCR base checked against the scan page by page); both quoted passages were additionally read off the page images before the footnotes were set, and the original's printed typos are reproduced verbatim. BibTeX entries generated from the frozen Scopus export with corpus/make_bib.py. Human-voice pass applied, including the paraphrase check against every footnote passage. Text reviewed and edited by the author before inclusion.

Section	Tool	Description
Theory (Sections 2.1–2.5)	Claude Code (Anthropic) + Gemini CLI (Google)	Blind citation check of the finished theory chapter (28 July 2026), continuing the dual-AI principle from the coding phase. Claude wrote the script corpus/gemini_verify_citations.py, which parses every citation site in a section and splits the check by what can be decided mechanically: whether a quote appears verbatim on the cited page is settled deterministically by the local retrieval tool, the source-echo rule (three or more consecutive words shared between the prose and the cited passage) is computed by the script, and only the judgement of paraphrase fidelity goes to Gemini 2.5 Pro — blind, receiving nothing but the sentence and the footnote passage. Of 25 citation sites in the chapter, the script flagged 7 for shared word sequences; the verdicts are recorded per site and collected in a report. The author decides every flag; the script proposes nothing and changes no text.

Section	Tool	Description
Method / bibliography	Claude Code (Anthropic)	Citation rule for the seven studies without printed page numbers (28 July 2026): each of the seven checked against its publisher type and the rule approved by the author implemented — article number in the bib eid field plus the article-internal page for the three Elsevier articles, section references for the four with publisher-relative or not-yet-assigned pagination. Two defects in bib.bib were found and fixed in the process: an unescaped ampersand in one title (would have aborted the LaTeX run at the first citation of it) and acronyms being lowercased by APA sentence casing, which had printed "ai", "sme" and "b2b" in the reference list. corpus/make_bib.py now escapes ampersands and brace-protects acronyms, with self-tests; the 27 existing entries were corrected in one pass and the reference list checked in the compiled PDF. One Scopus metadata error corrected against the source PDF: the author of the European Journal of Marketing article was recorded as "N., N. A." and is Tehrani, A. N.
Results (descriptive mapping)	Claude Code (Anthropic)	Drafting of Section 4.1 (18 July 2026): Claude drafted the descriptive-mapping prose directly from the finalized coding table (aggregate counts only, no source citations) and applied the agreed human-voice pass (banned-pattern audit, voice matching). All numbers cross-checked against the coding table; text reviewed and edited by the author before inclusion.